

SET- 5012

PGCP-BDA (Feb 26 Batch)

Date : 17/04/2026

**Note: Submit a .sql file for MYSQL with its name as your PRN number_name
(eg: 01_Rahul_Sinha.sql)**

Section 1: Basic SQL (6 Marks)

Q1. (6 Marks)

Write a CREATE TABLE statement for a table task with the following structure:

- task_id (Primary Key, Auto Increment)
- task_name (VARCHAR, NOT NULL)
- due_date (DATE)

Ensure the storage engine is InnoDB.

Insert the following records (let PK auto-generate):

- ('Data Cleaning', '2025-03-01')
- ('Model Training', '2025-03-15')

Update:

- Change task_name = 'Model Training' to 'ML Training' using a proper condition.

Retrieve all tasks where task_name starts with 'D'.

Section 2: Work on Given Schema (classicmodels) (34 Marks)

Q2. DDL & DML Operations (2 Marks)

a) Add a column:

profit_margin DECIMAL(5,2) NOT NULL DEFAULT 0.00
to the products table.

Then update it to 20.00 for all products where productLine = 'Classic Cars'.

b) Perform the following on customers table:

- Add column: contact_name VARCHAR(120)
- Populate using CONCAT(contactFirstName, ' ', contactLastName)

Q3. Aggregation & Joins (4 Marks)

List the following for all orders placed in February 2005:

- orderNumber
- customer_name (use CONCAT)
- total_quantity
- total_amount

Order the result by total_quantity (descending).

Q4. Analytical Query (3 Marks)

Find the top 5 products by total quantity sold.

Display:

- productCode
- productName

- total_quantity

Break ties using higher revenue.

Q5. Trigger (6 Marks)

Create a BEFORE UPDATE trigger on products such that:

- If MSRP < buyPrice, automatically set MSRP = buyPrice * 1.10
-

Q6. Stored Procedure (5 Marks)

Write a stored procedure:

get_customer_orders(p_customer_id INT)

It should return:

- orderNumber
- orderDate
- status
- total_amount (calculated)

Sorted by most recent orders first

Also show how to call it for:

customerNumber = 112

Q7. Function (5 Marks)

Create a deterministic function:

fn_total_quantity(p_orderNumber INT)

Return total quantity using:

SUM(quantityOrdered)

Demonstrate usage:

SELECT orderNumber, fn_total_quantity(orderNumber)

FROM orders;

Q8. CTE + Window Function (5 Marks)

Using a CTE + Window Function, rank customers by total payments made.

Output:

- customerNumber
- customer_name
- total_payment
- rank

Rank 1 = highest payment

Q9. (4 Marks)

Write a query to display:

- orderNumber
- orderDate
- customer_name
- total_products

Only include orders having total quantity greater than 100

Order result by orderDate.